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Quantitative Interpretation of NMR Relaxation Data

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Least-squares and linear programming algorithms for the interpretation of NMR relaxation data in terms of a spectrum of relaxation times are presented. These stable methods are noniterative and avoid all of the difficulties associated with nonlinear optimization schemes. Moreover, they are not restricted by any assumptions which limit the number of possible relaxation times. The inherent nonuniqueness due to the finite number of inaccurate measurements means that a variety of discrete and continuous spectra exist which fit the data equally well. For a complete interpretation the range of possible solutions must be examined. The algorithms discussed explore the space of acceptable solutions by constructing relaxation spectra from a continuum of models ranging from a few isolated delta functions to a continuous distribution. Alternatively, the interpreter can choose that class of spectrum (discrete or continuous) which best represents the physical system under investigation and so enhance the reliability of the interpretation. This flexibility is essential in order to avoid the pitfalls of interpreting a single spectrum returned by a single algorithm. In addition, the linear programming approach reduces the nonuniqueness by constructing unique information in the form of bounds on all spectra which satisfy the observations. These bounds and the corresponding extreme models are necessary for any quantitative determination of how spectral peaks change in amplitude or position with a change in the physical system. © 1989 Academic Press, Inc.

The investigation of NMR relaxation in inhomogeneous systems often involves the characterization of decay curves as a sum of exponential functions. For a single exponential decay or the sum of two exponentials with very different time constants, the estimation of T_1 or T_2 is relatively accurate. Unfortunately, for more complex systems which exhibit multiexponential relaxation, the estimation of time constants is a nontrivial problem. Nevertheless, the quantitative interpretation of NMR relaxation data in terms of a physical model requires a precise determination of the "true" distribution of relaxation times present in the sample.

The motivation for this study comes from work by our group on proton magnetic resonance of wood (*1*). The T_1 and T_2 values for water in wood cell lumens depend

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upon the diameter of the cylindrical cells (*2*). Since there is generally a distribution of cell diameters, wood samples should exhibit a distribution of T_1 and T_2 values. We therefore require reliable analyses of inversion-recovery and CPMG decay curves in order to understand the physical mechanism for the relaxation of water in wood. Similar situations arise with many other biological systems. The quantitative interpretation of relaxation data also has important applications in diagnostic medical magnetic resonance imaging.

The most common approach to the analysis of multiexponential relaxation is to fit the decay curves with the smallest number of discrete exponential terms which provide a satisfactory representation of the experimental data. For a variety of reasons, this approach becomes inaccurate when there are more than two or three exponential components or when any two time constants are similar. Another approach is to analyze the data in terms of a continuous distribution of relaxation times. This has been done both as a nonlinear problem by, for example, Provencher (*3*) and Smith and Nichols (*4*), and as a linear problem by Provencher (*5, 6*), Kroeker and Henkelman (*7*), and Kroeker, Stewart, Bronskill, and Henkelman (*8*). The continuous approach has the very desirable advantage of requiring few a priori assumptions about the relaxation behavior and it usually provides a more stable fit to the experimental data.

The aim of this paper is to discuss critically the application of linear inverse techniques to characterize decay curves in terms of discrete or continuous distributions of relaxation times. In particular, we investigate the nonuniqueness inherent in this problem. We demonstrate that a wide range of statistically acceptable, but quantitatively very different, solutions exist for multiexponential decay curves. We present algorithms which explore this space of acceptable models and appraise the nonuniqueness by finding bounds on spectra over relaxation time intervals.

To illustrate the application of these techniques, we use simulated decay curves from both continuous and discrete relaxation time distributions. We also analyze an actual CPMG decay curve from a western red cedar sapwood sample.

THEORY

The general integral equation describing multiexponential relaxation is

$$y(t_i) = y_i = \int_a^b s(T)e^{-t_i/T}dT, \qquad i = 1, 2, \ldots, N, \qquad [1]$$

where the N data y_i are measured at times t_i, and $s(T)$ is the unknown amplitude of the spectral component at relaxation time T. T represents either T_1 or T_2. The limits on the integral are restricted to the finite values a and b which are chosen to contain the values of T expected for the physical system being analyzed. The kernels of the equation are $e^{-t_i/T}$ and they are the amount a signal with relaxation time T has decayed by measurement time t_i. Equation [1] is easily generalized to have different kernel functions. For example, to solve for diffusion coefficients D using the pulsed field gradient method (*9*), the kernels are $e^{-\gamma^2 D\delta^2(\Delta-\delta/3)G_i^2}$, where γ is the gyromagnetic ratio, Δ and δ are the spacing and width of the field gradient pulses, and G_i are the gradient amplitudes. Alternatively, a spectrum of M_2 (interpair) values (*10*) may

be found using the kernels $e^{-M_{2i}\tau^2/2}$, where τ is the spacing between the two pulses of a solid echo sequence. In fact, any linear functional of a spectrum has the form of Eq. [1] and may be solved by the algorithms given below.

Equation [1] is a Fredholm integral equation of the first kind. Backus and Gilbert (*11–13*), Jackson (*14*), Lawson and Hanson (*15*), Parker (*16*), Twomey (*17*), Menke (*18*), and Oldenburg (*19*) discussed linear inverse theory solutions of such equations. These techniques are not iterative and do not require any initial estimate of the solution. In contrast, nonlinear optimization methods which solve Eq. [1] for amplitudes s_j at unknown times T_j often require a starting model to begin iterations. Nonlinear schemes are inappropriate because of the inherent difficulties of nonconvergence and convergence to only a local minimum. To reduce the effects of such problems, nonlinear techniques often employ the overly restrictive assumption of simply a few relaxation times. Linear inverse methods have no such difficulties or restrictions. These approaches assume a large number of known times T_j and solve for the corresponding amplitudes s_j, some of which may be zero. This is the approach taken by all of the algorithms discussed below.

For computer implementation, Eq. [1] must be discretized. The simplest and fastest method assumes that the spectrum is a sum of M delta functions with unknown areas s_j at known relaxation times T_j. That is,

$$s(T) = \sum_{j=1}^{M} s_j \delta(T - T_j). \qquad [2]$$

M is set large enough so as not to bias the solution into a small number of relaxation times. Typically M is between 100 and 200. Substituting Eq. [2] into [1] gives

$$y_i = \sum_{j=1}^{M} s_j e^{-t_i/T_j}, \qquad i = 1, 2, \ldots, N. \qquad [3]$$

Other discretizations are possible. For instance, if $s(T)$ is assumed to be piecewise constant over many small ranges of T then

$$y_i = \sum_{j=1}^{M} s_j g_{ij}, \qquad i = 1, 2, \ldots, N, \qquad [4]$$

where

$$g_{ij} = \int_{T_j}^{T_{j+1}} e^{-t_i/T} dT.$$

These g_{ij} are given analytically in terms of exponential integrals.

Equations [3] and [4] are linear systems of equations whose general form is

$$y_i = \sum_{j=1}^{M} A_{ij} s_j, \qquad i = 1, 2, \ldots, N. \qquad [5]$$

Additional linear constraints on the spectrum from other sources are incorporated as rows of the matrix **A**. Additional unknown variables to be found (such as the baseline offset, if any) require extra columns; however, in both cases the form of Eq. [5] is unchanged. This is the basic system solved by all of the following algorithms.

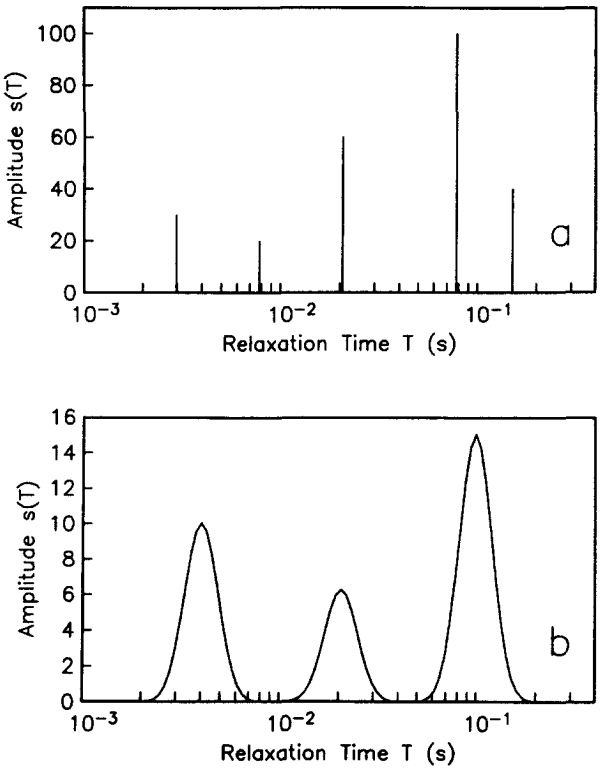


FIG. 1. Synthetic spectra. Synthetic spectra from two different classes are used to test the algorithms. A discrete spectrum of relaxation times is shown in (a). A spectrum with a continuous distribution of times is shown in (b).

All algorithms are tested on data corresponding to the two synthetic relaxation spectra illustrated in Fig. 1. Figure 1a shows a discrete spectrum composed of delta functions at a few isolated relaxation times. Figure 1b shows a spectrum with a continuous distribution of relaxation times which is representative of spin–spin relaxation of water in wood. Both spectra have the same total integral equal to 250 arbitrary units. Data corresponding to each spectrum were generated at $N = 100$ times t_i, logarithmically spaced between 0.001 and 0.500 s. Note that this sampling is a natural choice for exponential decays but that any other selection is possible. Backus and Gilbert (*13*) discussed methods which determine the sampling appropriate for optimal resolution of spectral features. The accurate data are plotted in Fig. 2 as solid curves. In any realizable experiment, the data y_i are corrupted by noise. It is assumed here that the noise is uncorrelated, zero mean, and Gaussian with a standard deviation of σ_i for the ith datum. A realization of such noise with $\sigma_i = 0.5$ units was added to the accurate data. The resulting noisy data are plotted as symbols in Fig. 2. The error bars indicate one standard deviation.

Because of the noise contaminating y_i, Eq. [5] must not be solved exactly. Proposed solutions must misfit the inaccurate data by an amount consistent with the errors. The χ^2 statistic quantifies the misfit between the measured data

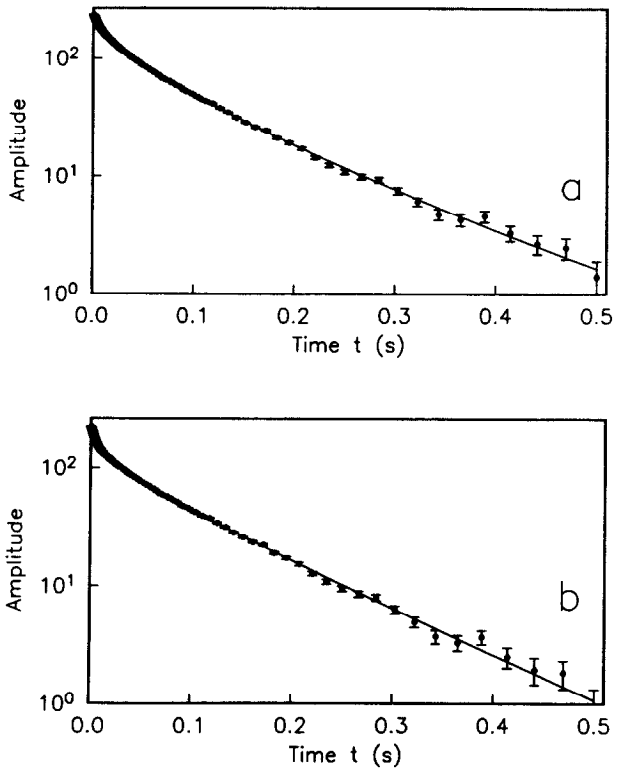


FIG. 2. Synthetic data. The solid lines in (a) and (b) are accurate data from the test models in Figs. 1a and 1b, respectively. The 100 circles in each frame are logarithmically spaced, noisy data. These data are derived from the accurate data (solid lines) by adding uncorrelated, zero mean, Gaussian noise. The error bars indicate plus or minus one standard deviation. These noisy data are used as input to all the algorithms.

y_i and the data y_i^p corresponding to a particular constructed model $s(T)$. The χ^2 misfit is

$$\chi^2 = \sum_{i=1}^{N} (y_i^p - y_i)^2/\sigma_i^2. \qquad [6]$$

The preferred model has $\chi^2 \simeq N$, which is the expected value of the χ^2 statistic. In this case, each datum is misfit by about one standard deviation. If $\chi^2 \ll N$, the data are fit too accurately and structure that is merely an artifact of the noise will be evident on $s(T)$. Alternatively, if $\chi^2 \gg N$, the data are not fit closely enough and information about the true spectrum contained in the data will be lost.

Given the noisy data in Fig. 2 (symbols), the expected value of χ^2 is 100. Some maximum value of χ^2 is defined as the boundary between acceptable and unacceptable models. In this paper, the upper limit is fixed at 109. From the statistics of the χ^2 distribution this is the 75% confidence limit. Thus, a constructed model with $\chi^2 = 109$ could be the true model and there is a 25% chance that the noise alone has caused the measurements y_i to differ so much from the data y_i^p corresponding to the true $s(T)$. In other words, only 25% of noise realizations are sufficiently erratic such

that, when added to the true data, χ^2 is greater than or equal to 109. As an example of how χ^2 varies with the confidence limit, note that for $N = 100$, the 10% confidence limit is 118, and the 5% limit is 124. Other misfit criteria may be used to alter somewhat the boundary between acceptable and unacceptable models. For the example calculations, $M = 100$ relaxation times T_j were selected, logarithmically spaced between 0.001 and 0.400 s. Again, this is an arbitrary choice, and a linear or irregular sampling would do as well.

The algorithms discussed here fall into two groups. The first group contains least-squares approaches similar to those developed by Provencher (*5, 6*) and used by Kroeker and Henkelman (*7*). The second group contains new linear programming methods.

LEAST-SQUARES APPROACHES

Nonnegative least-squares: NNLS. The NNLS algorithm of Lawson and Hanson (*15*) is ideally suited to relaxation time analysis. The method solves Eq. [5] using stable Householder transformations and is guaranteed to find a solution in a finite number of these operations. The nonnegativity of the spectral amplitudes s_j is implicit in the formulation and need not be included as extra rows in the constraint matrix **A**. NNLS finds that set of s_j which minimizes the least-squares misfit

$$\sum_{i=1}^{N} \left| \sum_{j=1}^{M} A_{ij} s_j - y_i \right|^2. \qquad [7]$$

Because of the noise contaminating the measurements y_i it is not possible to reduce this misfit to zero.

In practice each datum and its corresponding row of **A** are first divided by the standard deviation σ_i. This normalizes the data to unit variance and reduces the influence of very inaccurate data on the solution. With this normalization the least-squares solution constructed by NNLS minimizes the χ^2 misfit of Eq. [6]. Note that if the least-squares solution has $\chi^2 \gg N$ then no $s(T)$ model exists which adequately reproduces the data. In this case, the physical system is exhibiting nonexponential relaxation and the governing Eq. [1] must be adapted accordingly.

Figure 3 shows the output from NNLS for the two noisy data sets of Fig. 2 (symbols). The delta-function parameterization of Eq. [2] is used for all examples in this paper. In both panels the true spectrum is a dashed line and the constructed spectrum is a solid line. Figure 3a shows the discrete case where the NNLS model has $\chi^2 = 79$. The algorithm places delta functions near the correct times of $T = 0.003$, 0.008, 0.021, 0.078, and 0.152 s. The one at 0.021 s is almost exactly correct but all other peaks show shifts in times and amplitudes. This is the best-fitting spectrum in a least-squares sense given the noisy data and the partition of the relaxation time axis. Note that the χ^2 misfit of the true discrete model to the noisy data is 91 which is larger than the minimum possible. Thus the true model is bypassed in the descent to the minimum χ^2. Using accurate data results in a least-squares solution indistinguishable from the true model of Fig. 1a.

NNLS models tend to be composed of a few isolated delta functions. When the true spectrum also has this form then the agreement with the constructed model is

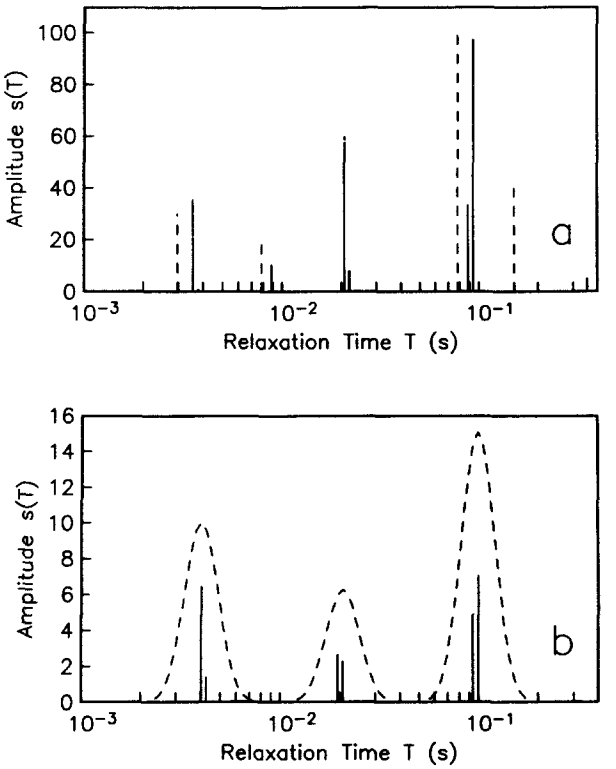


FIG. 3. NNLS least-squares spectra. The solid lines in (a) and (b) are NNLS spectra constructed using the noisy data of Figs. 2a and 2b (symbols), respectively. The solid line in (b) has been normalized by dividing by 10. The dashed lines are the true spectra from Figs. 1a and 1b. This figure illustrates how delta-function models are able to fit data from discrete or continuous spectra.

good. However, it is possible for the physical system under investigation to have a continuous distribution of relaxation times. Figure 3b shows a spectrum from this class (dashed line) and the least-squares model (solid line) derived from the corresponding noisy data of Fig. 2b. The χ^2 misfit is 80. The discrete spectrum amplitudes have been divided by 10 in order to fit within the frame. The delta functions are located near the proper relaxation times and their integral over a local range is approximately equal to the integral of the true model over a similar range. Figure 3b illustrates the nonuniqueness of any inverse problem typified by equations such as [1]. In this case, a delta-function model reproduces the observations very well but it has a completely different character than the true spectrum which generated the data. This difference in form persists even when 100 accurate data are used and NNLS constructs a model which exactly reproduces the data.

Fortunately, NNLS is easily adapted to construct a continuous spectrum which may be a more realistic model of the particular physical system under investigation. Extra constraints are incorporated into the matrix **A** of Eq. [5] to alter the discrete character of the basic least-squares solution. A general form is to minimize

$$\sum_{i=1}^{N} \left| \sum_{j=1}^{M} A_{ij}s_j - y_i \right|^2 + \mu \sum_{k=1}^{K} \left| \sum_{j=1}^{M} H_{kj}s_j - f_k \right|^2 \quad [8]$$

for fixed $\mu > 0$, where H_{kj} is a matrix representing K additional constraints, and f_k is the corresponding vector of right-hand side values. The least-squares solution is obtained when the trade-off parameter $\mu = 0$. The larger the value of μ the more the algorithm attempts to satisfy the constraints involving the matrix **H** at the expense of increasing the χ^2 misfit. The value of χ^2 increases monotonically with μ and in the limit of large μ the data equations are completely ignored. The value of μ is adjusted until the misfit is near its expected value of N, which results in the preferred model in the absence of other information.

The matrix **H** can take many forms. If **H** is the $M \times M$ identity matrix and **f** is the zero vector, then the $K = M$ extra constraints minimize the "energy" in the spectrum:

$$\sum_{j=1}^{M} s_j^2. \quad [9]$$

If **f** is nonzero, then the constraints minimize the difference between the constructed spectrum and the a priori estimate **f**. Another possible form of **H** is to set the rows equal to finite difference approximations of first-order or second-order derivatives of $s(T)$. When **f** is zero, these cases respectively minimize the energy in the spectrum derivative

$$\sum_{j-1}^{M-1} |s_{j+1} - s_j|^2, \quad [10]$$

or curvature

$$\sum_{j=1}^{M-2} |s_{j+2} - 2s_{j+1} + s_j|^2. \quad [11]$$

Provencher (*5*) chose the second-derivative formulation to construct smooth $s(T)$. For noisy data, additional constraints of the form of Eqs. [9], [10], or [11] are essential to regularize the solution. This is analogous to adding a ridge regression parameter to stabilize a matrix inversion in the presence of noise.

Figure 4a is an example of a least-squares minimization of Eq. [8] with the $K = M - 1$ first-derivative constraints of Eq. [10]. The smooth spectrum calculated (solid line) has $\chi^2 = 103$. The true spectrum (dashed line) is discrete and has been normalized by 10 to fit within the frame. The first-derivative constraints smooth out the amplitudes of the NNLS discrete solution while increasing the misfit to a more appropriate value closer to N. In Fig. 4b, the constructed and true spectrum (solid and dashed lines, respectively) are both continuous and so there is a much better agreement in the overall shapes. The constructed model has broader and lower peaks but its total integral is the same as that of the true model. Reducing the value of μ will narrow the peaks and increase their amplitude while decreasing χ^2. On the basis of χ^2 alone the two models plotted in Fig. 4b are virtually indistinguishable because the true spectrum (dashed line) has $\chi^2 = 91$ and the constructed model misfit is 95. Figure 4 demonstrates that a smooth profile can represent data from a discrete or continuous system equally well.

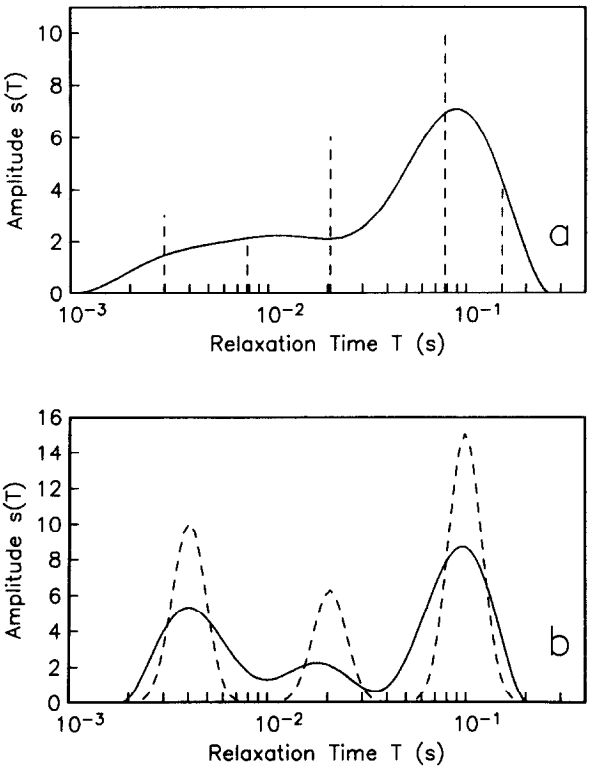


FIG. 4. NNLS spectra with first-derivative constraints. The solid lines in (a) and (b) are NNLS spectra which minimize a combination of misfit and model smoothness. The dashed lines are the true spectra and that in (a) has been normalized by 10. This figure complements Fig. 3 and shows how smooth models fit data from discrete or continuous spectra.

Least-distance programming: LDP. The LDP algorithm of Lawson and Hanson (*15*) is also applicable to relaxation time analysis. This method is for problems with fewer data than unknown spectral amplitudes, that is, $N < M$. This situation typically arises for T_1 analyses. Such underdetermined systems have no unique solution because many spectra satisfy the limited number of inaccurate data. One solution must be chosen out of the infinite number possible. A minimum structure profile is a good choice because it tends to have the essential features required by the data and no spurious structure to mislead the interpreter (*5, 16, 20*). NNLS generates minimum structure models by minimizing Eq. [7] to find a few delta functions which fit the data well, or by minimizing Eq. [8] which is a combination of misfit and minimum structure. On the other hand, LDP selects that model which minimizes Eq. [9] directly, subject to fitting the data. This results in the minimum energy or least-distance solution for a given misfit. The data constraints are imposed as the $2N$ inequalities

$$y_i - a\sigma_i \leq \sum_{j=1}^{M} A_{ij}s_j \leq y_i + a\sigma_i, \qquad i = 1, 2, \ldots, N, \qquad [12]$$

where a is a positive parameter governing the misfit. Setting $a \simeq 2$ usually results in

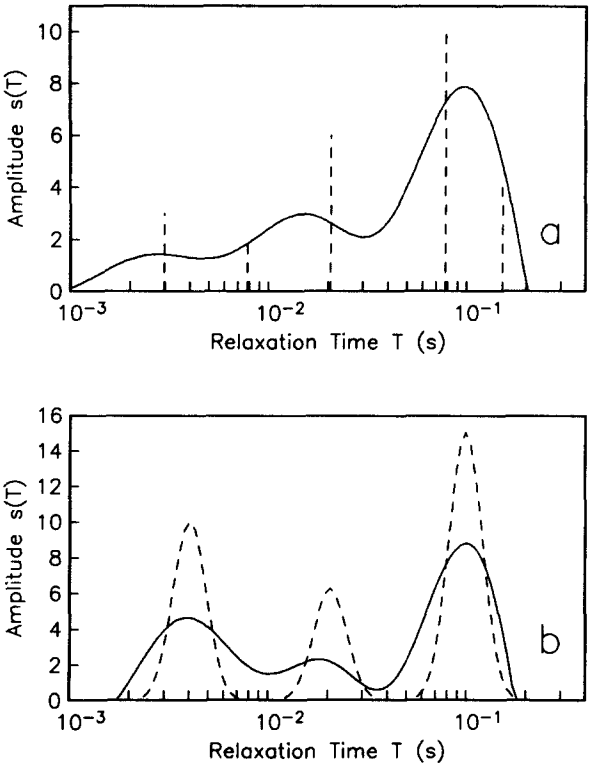


FIG. 5. LDP least-distance programming spectra. The solid lines in (a) and (b) are LDP spectra. The dashed lines are the true spectra and that in (a) has been normalized by 10.

models with χ^2 near the expected value of N. Positivity constraints must also be included. These take the form

$$\mathbf{Is} \geqslant \mathbf{0},$$

where $\mathbf{I}$ is the $M \times M$ identity matrix, $\mathbf{s}$ is the $M \times 1$ vector of spectrum amplitudes, and $\mathbf{0}$ is the $M \times 1$ zero vector. Positivity of the solution adds M extra constraints to the $2N$ of Eq. [12]. Any additional inequality constraints such as upper or lower bounds on linear combinations of s_j may also be imposed.

Figure 5 shows LDP spectra constructed from the noisy data sets. Their character is very similar to the NNLS solutions with first-derivative constraints shown in Fig. 4. The χ^2 misfits are 99 and 101 for Figs. 5a and 5b, respectively. The true discrete model in Fig. 5a (dashed line) has been normalized by 10 to fit within the frame. Because these are minimum energy solutions, the peaks probably correspond to amplitudes in the true spectra rather than being simply artifacts of the noise or the inversion scheme. Thus, for example, the noisy data of Fig. 2b likely require spectra with significant amplitudes near $T = 0.004$, 0.020, and 0.100 s.

The algorithms NNLS and LDP construct models from opposite ends of a continuum of possible solutions. NNLS begins with best-fitting delta-function models. It

constructs smoother models when the misfit is allowed to increase and constraints such as Eqs. [9], [10], or [11] are imposed. In contrast, LDP finds the model with the smallest possible energy at some specified misfit governed by the parameter a. It constructs less smooth models as a decreases and the misfit is forced toward the minimum possible value given by NNLS. Note that for overdetermined problems with $N > M$, LDP will still return smooth models as long as N is not too much larger than M.

LINEAR PROGRAMMING APPROACHES

Model construction. Linear programming (LP) is a different approach to solving a linear system of equations (*21, 22*). LP finds the solution which satisfies the data constraints of Eq. [12] and which minimizes or maximizes the L_1-norm objective function

$$\sum_{j=1}^{M} w_j s_j, \qquad [13]$$

where w_j are arbitrary weights. Nonnegativity of the spectrum amplitudes s_j is implicit in the LP formulation. Moreover, LP easily incorporates any additional equality or inequality constraints on linear combinations of s_j that may come from sources external to the given experiment. As with the least-squares algorithms, these may be upper and lower bounds, smoothness constraints, or a priori estimates.

LP solutions tend to have a small number of isolated delta functions similar to the least-squares version of NNLS; however, the misfit is adjustable via the parameter a in Eq. [12] so that LP can construct more than just the one best-fitting model. If the physical system is expected to be composed of a few isolated delta functions then LP can generate such a model with the appropriate χ^2 misfit.

Figure 6 shows an example of LP construction. The objective function of Eq. [13] with all $w_j = 1$ was minimized and the misfit parameter a of Eq. [12] was set at 2.0. The models resemble the NNLS least-squares solutions in Fig. 3 but the misfits are higher and at the appropriate level, being 105 and 102 for Figs. 6a and 6b, respectively. The constructed model in Fig. 6b (solid line) has been normalized by 10.

Figure 6 demonstrates that the basic LP approach is suited to interpreting physical systems which are discrete. However, the LP method can also construct smooth profiles. Including the $M - 1$ extra inequality constraints of the form

$$|s_{j+1} - s_j| \leqslant \beta, \qquad [14]$$

where $\beta > 0$, restricts adjacent solution amplitudes to be of similar value. The minimum possible value of β results in the most uniform set of s_j consistent with the data constraints. A smaller β results in an inconsistent system of equations with no solution. It is also possible to reformulate the LP problem and change from the local constraints of Eq. [14] to a global bound on the total variation of the spectrum (*23*).

Figure 7 shows LP spectra with the constraints of Eq. [14] added. For these examples $\beta = 2.0$. The true model in Fig. 7a (dashed line) has been normalized by 10. The χ^2 misfits are 108 and 100 for the constructed models (solid lines) in Figs. 7a and 7b, respectively. These spectra are more peaked than those from the least-squares

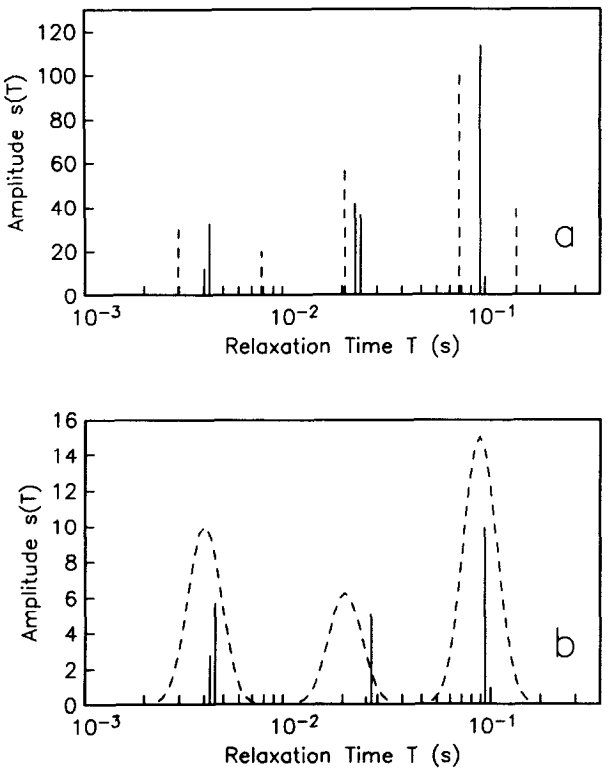


FIG. 6. LP linear programming spectra. The solid lines in (a) and (b) are LP spectra. The solid line in (b) has been normalized by 10. The dashed lines are the true spectra.

algorithms plotted in Figs. 4 and 5. This is because the absolute value L_1-norm of LP is influenced less by sharp changes in the model than the least-squares L_2-norm of Eq. [10] which discriminates strongly against large gradients.

Model appraisal. The spectra plotted in the preceding figures illustrate the nonuniqueness of relaxation time analysis. Many models adequately fit the data, and chosing the one most likely to represent reality is difficult. Figures 4, 5, and 7 show how smooth profiles can fit data from smooth as well as discrete systems. Figures 3 and 6 demonstrate equally good fits for the completely different class of discrete models. Using physical arguments to constrain the solution to be from either the discrete or the continuous class of functions reduces the nonuniqueness; however, within each class there remain substantial variations in peak amplitudes and locations. Appraisal techniques (*13, 19*) are necessary to explore the extent of the nonuniqueness and avoid the pitfalls of interpreting simply a single spectrum constructed by a single algorithm.

One way of examining the nonuniqueness is to use many different algorithms to construct diverse, minimum structure models which fit the data. Figures 3 through 7 illustrate the results from NNLS, LDP, and LP discrete and smooth interpretations. Features common to all profiles are likely to be required by the data and not just artifacts of the noise or the inversion method. However, this interpretation can never

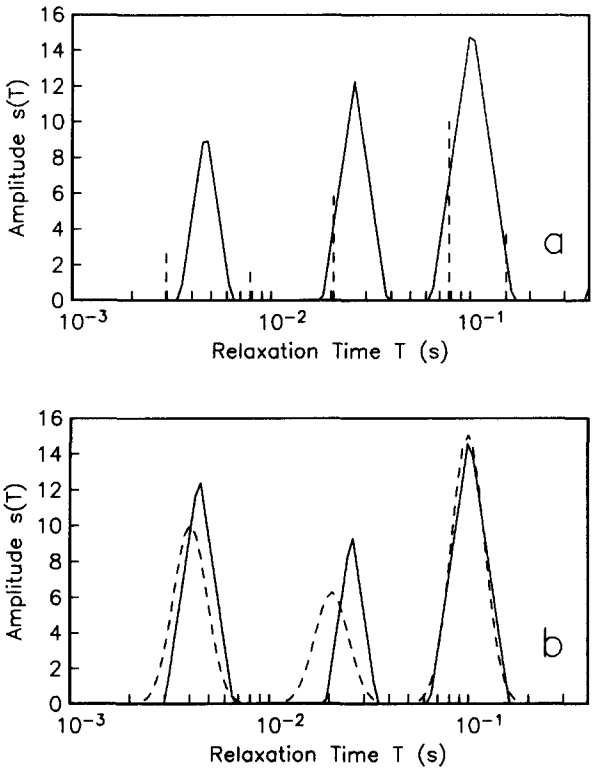


FIG. 7. LP spectra with first-derivative constraints. The solid lines in (a) and (b) are LP spectra with first-derivative constraints added. The dashed lines are the true spectra and that in (a) has been normalized by 10. The peaks are sharper than those of Figs. 4 and 5 because the least absolute value constraints of Eq. [14] are less strict than the least-squares constraints of Eq. [10].

be completely quantitative because no combination of construction methods can ever find the infinite number of diverse models which satisfy the data.

Fortunately, LP methods can make quantitative statements about the nonuniqueness. The L_1-norm objective function in Eq. [13] provides an extra dimension of flexibility to find spectra that fit the data and have specific characteristics. The weights w_j may be adjusted to minimize or maximize any linear combination of the amplitudes s_j. For example, uniform weights result in models which provide the interpreter with upper and lower bounds on the integral of $s(T)$ which is the total amount of signal allowed by the measured data. Alternatively, some weights may be set to zero to construct profiles which minimize or maximize the L_1-norm of the spectrum within localized relaxation time intervals.

Figure 8 gives an example of solutions involving a nonuniform objective function. The task is to quantify the bounds on the fraction of spectral amplitude within localized ranges in T. For this example, the ranges are 0.001 to 0.010 s, 0.010 to 0.050 s, and 0.050 to 0.400 s. To solve this problem the objective function weights w_j are set to unity within the first of these intervals and set to zero elsewhere. Then two solutions satisfying the data are found, one which minimizes the objective function and the

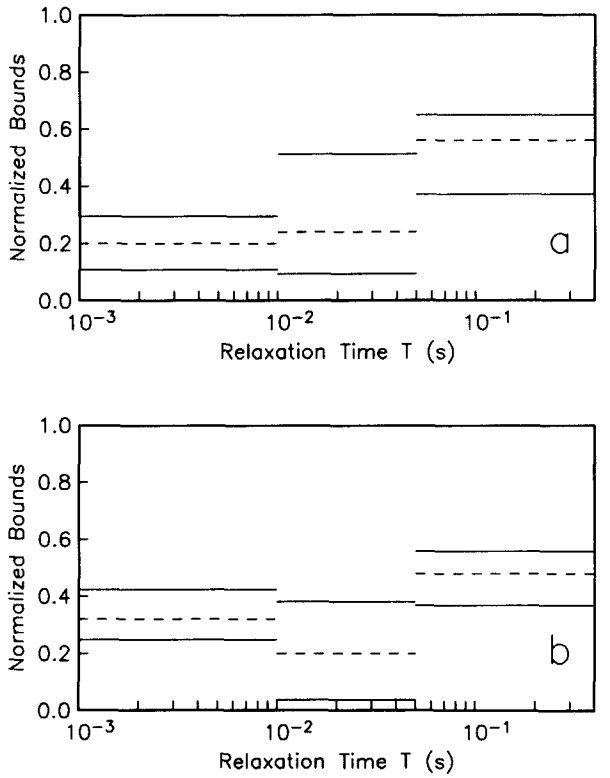


FIG. 8. LP localized bounds. The solid lines in (a) and (b) are LP bounds on the integrated spectral amplitudes within each of three intervals, normalized by the total integral of the spectrum. The discrete spectrum data were used in (a), and the continuous spectrum data were used in (b). The ranges are from 0.001 to 0.010 s, from 0.010 to 0.050 s, and from 0.050 to 0.400 s. The dashed lines are the integrals of the true spectra.

other which maximizes it. Repeating this procedure for the remaining two ranges gives a total of six models which minimize or maximize the spectrum amplitude within each of the three intervals independently. Figures 8a and 8b give the bounds (solid lines) for the discrete and continuous spectrum data, respectively. These levels correspond to the integrated amplitude within the range divided by the total integrated amplitude of the model. The dashed lines are the amplitude fractions of the true models in Fig. 1. These bounds communicate a sense of how much variation is possible over selected intervals. As the interval narrows, the bounds widen. The lower bound for very thin intervals is near zero because it is easy to move amplitudes to locations immediately outside the boundaries. The bounds meet for an interval spanning the full range of T values. Thus, the wider the interval the more the data constrain the integral of the spectrum within the interval.

These bounds are not approximations. Given the noisy data, a sampling of the T axis, and an acceptable χ^2 misfit level, these are absolute bounds on the set of all spectra having an equal or smaller misfit. This is a different way of representing the information in the data. Instead of constructing spectra which can have substantial

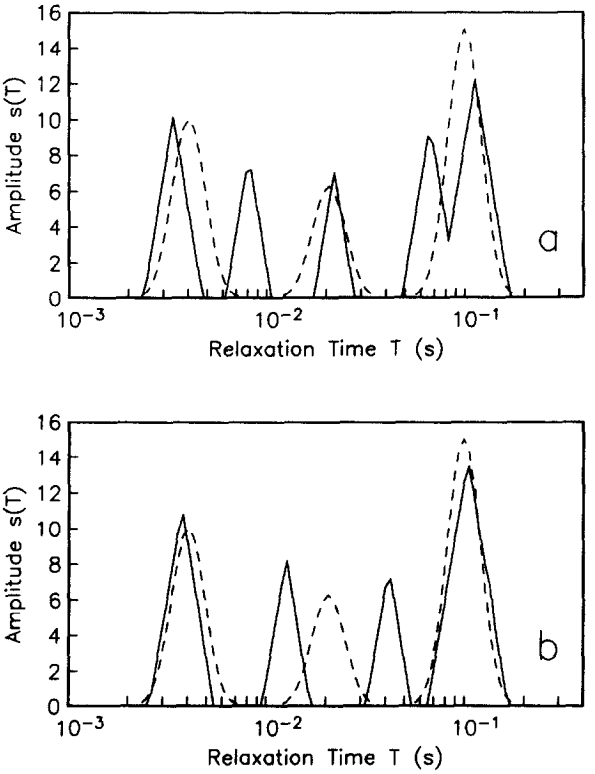


FIG. 9. Extreme LP spectra. The solid lines in (a) and (b) are LP spectra which minimize and maximize, respectively, the signal within the central interval from 0.010 to 0.050 s. The additional constraints of Eq. [14] are imposed with $\beta = 1.8$. The dashed lines are the true spectra. Despite fitting the data equally well, the two models represent the central interval in completely different ways.

variations in form (discrete versus continuous, for example), LP appraisal sacrifices the resolution of model details to gain quantitative knowledge common to all acceptable spectra of features spanning finite T intervals.

To demonstrate how the bounds are used, consider Fig. 8b. It is possible to construct spectra which adequately fit the 100 noisy data in Fig. 2b but which allow as little as 4% to as much as 38% of the signal to lie within the central interval. The actual percentages depend on the misfit permitted. A smaller misfit results in narrower bounds. For Fig. 8 all of the corresponding spectra had χ^2 between 96 and 104. Hence, considering solely the 100 noisy data it is not possible to classify the data as coming from one of two systems which have, say, 10 and 30% of their signal in this interval. Other criteria must be added in order to categorize the data on the basis of signal percentage within the central interval. For example, to narrow the bounds, Dosso and Oldenburg (*23*) limited the total variation of the model. Otherwise, the nonuniqueness inherent in this inverse problem is too great.

Figure 9 illustrates the nonuniqueness in the character of spectra which fit the smooth model data. The signal within the central interval was minimized and maximized using objective function weights as for Fig. 8. However, the additional con-

straints of Eq. [14] with $\beta = 1.8$ were also imposed to generate smooth profiles. Reducing the variation in this way narrows the bounds. The minimization solution shown in Fig. 9a (solid line) has 11% of the total amplitude within the central interval, whereas the maximization result in Fig. 9b (solid line) has 28%. These percentages are within the limits plotted in Fig. 8b. The χ^2 misfits are 97 and 101 for Figs. 9a and 9b, respectively. In both extreme cases, the true peaks centered on 0.004 and 0.100 s (dashed lines) are well reproduced; however, there are dramatic differences in the central peak. In Fig. 9a signal is placed just outside the central interval but a small peak near 0.020 s survives. In Fig. 9b the signal is placed just inside the interval and the true peak at 0.020 s is totally misrepresented. Figure 9 gives two spectra which fit the data equally well but which have a completely different character within the central interval. Thus interpretations based on the presence or absence of a central peak are untenable.

Bounds and extreme models such as plotted in Figs. 8 and 9 are essential for any quantitative analysis of relaxation time spectra. The variability and the extremes allowed by the limited number of inaccurate data must be examined in this way. Using a single algorithm to construct a single model does not provide sufficient information on which to base a responsible interpretation.

ANALYSIS OF EXPERIMENTAL DATA

Figure 10a shows the CPMG decay curve from a western red cedar (*Thuja plicata* Donn) sapwood sample with a moisture content of 228%. The moisture content is the ratio of the weight of water to that of wood. There are $N = 174$ data points. One standard deviation error bars are plotted as in Fig. 1 but the data are so accurate that the upper and lower limits are mostly indistinguishable on this scale. Figure 10b gives the NNLS T_2 least-squares spectrum (solid lines) corresponding to the data in Fig. 10a. The model amplitudes have been divided by 10. The least-squares χ^2 is 36, which is well below the expected value of 174. Incorporating the first-derivative constraints from Eq. [10] into an NNLS construction results in the continuous model shown in Fig. 10b (dashed line). The continuous spectrum misfit of 172 was adjusted to be near N by using the trade-off parameter μ.

The major signal near $T_2 = 0.200$ s appears on both models; however, the locations of amplitudes at smaller T_2 values are different for the discrete and continuous spectra. This is expected because of the large difference in misfits between these two models. The LP method can generate discrete spectra similar to those from the NNLS least-squares algorithm but with a more appropriate χ^2 closer to N. The LP method then gives discrete peaks (not shown) nearer to those in the continuous model.

The preferred model is the smooth curve in Fig. 10b (dashed line) because we expect that lumen water in wood possesses a continuous distribution of T_2 values corresponding to the continuous distribution of lumen diameters. The three peaks of the smooth spectrum correspond to three environments for water in wood. The peak at 0.006 s is due to water bound inside the cell walls, and the peaks at 0.030 and 0.200 s are due to water in cell lumens of smaller and larger diameters, respectively.

To investigate the persistence of the three peaks present in the continuous spectrum, the LP appraisal algorithm was run using three T_2 ranges: 0.001 to 0.012 s,

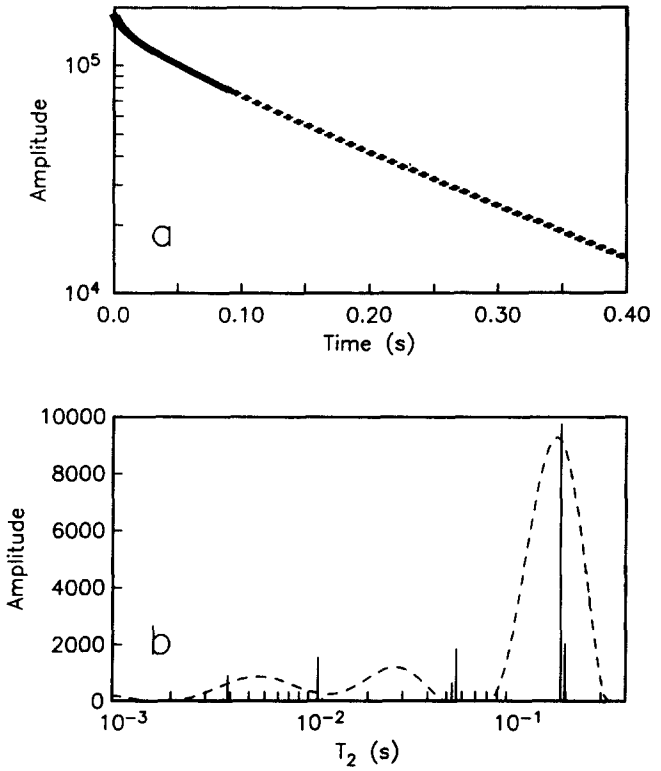


FIG. 10. Western red cedar CPMG data and NNLS spectra. The symbols in (a) are the 174 CPMG decay curve data from a western red cedar sapwood sample. The error bars indicate plus or minus one standard deviation but are difficult to resolve on this scale. The corresponding NNLS discrete and smooth spectra are given in (b) by the solid and dashed lines, respectively. The discrete spectrum amplitudes have been divided by 10.

0.012 to 0.075 s, and 0.075 to 0.400 s. Figure 11a shows the results. The average signal amplitude over the highest range near $T_2 = 0.200$ s is well constrained by the data to be at least 65% but no more than 82% of the total amplitude. There must also be at least 5% of the signal within the lowest T_2 range from 0.001 to 0.012 s. The central interval allows the most variation. The lower bound of zero indicates that the data do not require any signal at all over this central range. Localized bounds such as these are important for a quantitative determination of whether or not significant differences exist between the spectra from different wood species.

Figure 11b illustrates another way of using LP to aid an interpretation. Each pair of dots at a particular T_2 value represents the upper and lower bounds on the normalized integral of $s(T_2)$ up to that T_2 value. For example, the lower bounds demonstrate that the data do not require any signal below $T_2 = 0.006$ s. The upper bounds show that no additional amplitude is required above $T_2 = 0.200$ s. In between, the bounds show how most of the signal must be concentrated between 0.100 and 0.230 s. When wood is dried, the water leaves the larger lumens first, resulting in a loss of signal at longer T_2 values. The cumulative bounds determine the amount and location of the remaining water so that the drying process may be studied in detail.

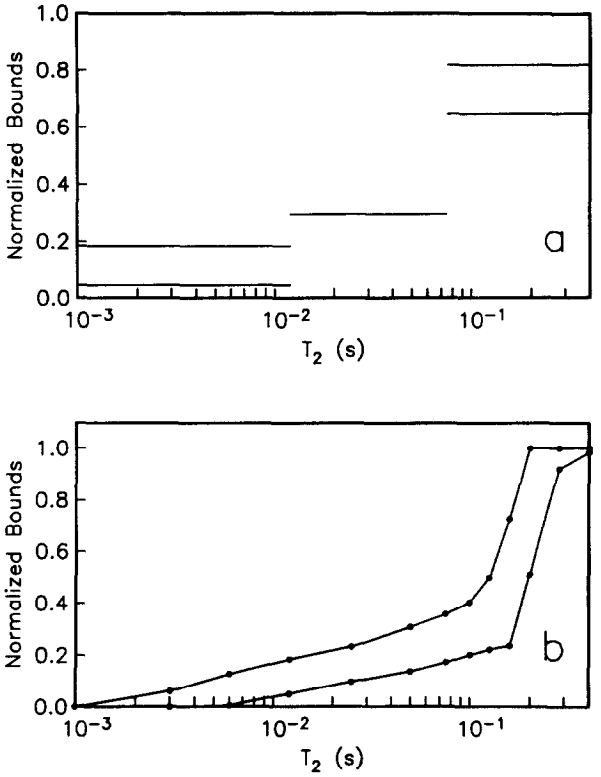


FIG. 11. Localized and cumulative LP bounds on cedar spectra. The pairs of lines in (a) are LP bounds on the integrated amplitudes within each of three intervals, normalized by the total integral of the cedar spectrum. The three ranges are from 0.001 to 0.012 s, from 0.012 to 0.075 s, and from 0.075 to 0.400 s. Each pair of dots in (b) at a particular T_2 value represents upper and lower bounds on the normalized integral of $s(T_2)$ up to that T_2 value.

CONCLUSIONS

The least-squares and linear programming algorithms presented here solve the linear integral Eq. [1] for a discrete or smooth spectrum of relaxation times $s(T)$ satisfying the data to within their errors. These stable methods require no starting model, no iterative solution, and no restrictive assumptions limiting the spectrum to a few delta functions. All the approaches generate minimum structure models, whether a few isolated delta functions or a smoothly varying profile. This philosophy of parsimony (*5, 16, 20*) prevents generation of spurious features not required by the data and so increases confidence in the resulting structure. Moreover, both methods can incorporate additional equality or inequality constraints on any linear combination of the spectral amplitudes.

These construction methods must be supplemented by techniques which appraise the extent of the nonuniqueness. Figures 4 and 6, for example, show that discrete or smooth models fit data from both discrete and smooth spectra equally well. Figure 9 demonstrates that restricting models to be smooth still allows substantial variation. Using the LP method to minimize and maximize the objective function generates different information in the form of bounds on the set of all spectra which fit the data.

This appraisal aspect of LP is not concerned with the features of any one model but with features common to all acceptable spectra. Examination of the spectra corresponding to the bounds provides further insight into the variability permitted by the data.

In addition, the NNLS algorithm answers the question of existence by finding that $s(T)$ with the minimum least-squares misfit. If the minimum value is unacceptably high then the relaxation mechanism is different or more complex than the equation being used to represent the physical system (Eq. [1], for example).

Existence and the construction and appraisal methods presented here depend on obtaining good estimates of the standard deviations σ_i. Extra experimental effort devoted to determining these values leads to rich rewards in the amount of information returned. Data without error bars are virtually meaningless because there is no measure of confidence in their values. This translates to no confidence in the determination of what possible variations are permitted by the data and which spectrum is optimal.

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